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Cooled chokes for ablation systems and methods of use

Abstract

An antenna system for tissue ablation comprises an energy transmission member, an antenna body coupled to the energy transmission member, a fluid source, and a choke member including a choke body and a choke connector electrically coupling the choke body to the energy transmission member. The choke connector is in direct contact with fluid from the fluid source and forms a delivery path for the fluid between the choke body and the energy transmission member. The choke connector and the choke body are integrally formed of a continuous wire mesh.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is the continuation of U.S. patent application Ser. No. 16/681,255 filed Nov. 12, 2019, which claims the benefit of U.S. Provisional Application No. 62/760,583, filed Nov. 13, 2018, both of which are incorporated by reference herein in their entirety.

FIELD

(1) The present disclosure is directed to minimally invasive ablation systems and methods of use and more particularly to systems and methods for cooling minimally invasive ablation systems and associated chokes.

BACKGROUND

(2) Minimally invasive medical techniques are intended to reduce the amount of tissue that is damaged during medical procedures, thereby reducing patient recovery time, discomfort, and harmful side effects. Such minimally invasive techniques may be performed through natural orifices in a patient anatomy or through one or more surgical incisions. Through these natural orifices or incisions, an operator may insert minimally invasive medical tools to reach a target tissue location. Minimally invasive medical tools include instruments such as therapeutic, diagnostic, biopsy, and surgical instruments. Minimally invasive medical tools may also include ablation instruments. Ablation instruments transmit energy in the form of electromagnetic waves to a targeted area of tissue, such as a tumor or other growth, within the patient anatomy to destroy the targeted tissue. Some minimally invasive medical tools and ablation instruments may be teleoperated or otherwise computer-assisted. Various features may improve the effectiveness of minimally invasive ablation instruments.

SUMMARY

(3) Embodiments of the invention are best summarized by the claims that follow the description.

(4) Consistent with some embodiments, an antenna system for tissue ablation may comprise an energy transmission member, an antenna body coupled to the energy transmission member, a fluid source, and a choke member including a choke body and a choke connector electrically coupling the choke body to the energy transmission member. The choke connector may be in direct contact with fluid from the fluid source and forms a delivery path for the fluid between the choke body and the energy transmission member. The choke connector and the choke body may be integrally formed of a continuous wire mesh.

(5) Consistent with some embodiments, a method for delivering ablation energy may comprise conducting energy through an energy transmission member and conducting energy through a choke member coupled to the energy transmission member. The choke member includes a choke body and a choke connector electrically coupling the choke body to the energy transmission member. The choke member may be formed from a continuous wire mesh. The method may further comprise providing fluid from a cooling system through the choke member such that the choke

connector is in direct contact with the fluid.

(6) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional aspects, features, and advantages of the present disclosure will be apparent to one skilled in the art from the following detailed description.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

- (1) FIG. 1A illustrates a partial view of an antenna assembly with a choke member according to some embodiments.
- (2) FIG. 1B illustrates a cross-sectional view of the antenna assembly of FIG. 1A according to some embodiments.
- (3) FIG. 2A illustrates a schematic view of fluid flow in an antenna system according to some embodiments.
- (4) FIG. 2B illustrates a schematic view of fluid flow in an antenna system according to some embodiments.
- (5) FIG. 2C illustrates a schematic view of fluid flow in an antenna system according to some embodiments.
- (6) FIGS. 3-9 illustrate partial views of antenna systems with choke members according to some embodiments.
- (7) FIG. 10 illustrates a method for forming an antenna choke member according to some embodiments.
- (8) FIGS. 11A-11D illustrates a sequence of views of a procedure for forming a choke member according to some embodiments.
- (9) FIGS. 12-15 illustrate a partial views of antenna systems with a choke member according to some embodiments.
- (10) FIG. 16 illustrates a method for transferring energy to an ablation site according to some embodiments.
- (11) FIG. 17 is a simplified diagram of a teleoperated medical system according to some embodiments.
- (12) FIG. 18A is a simplified diagram of a medical instrument system according to some embodiments.
- (13) FIG. 18B is a simplified diagram of a medical instrument with an extended medical tool according to some embodiments.
- (14) Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

(15) In the following description, specific details are set forth describing some embodiments consistent with the present disclosure. Numerous specific details are set forth in order to provide a thorough understanding of the embodiments. It will be apparent, however, to one skilled in the art that some embodiments may be practiced without some or all of these specific details. The specific embodiments disclosed herein are meant to be illustrative but not limiting. One skilled in the art may realize other elements that, although not specifically described here, are within the scope and the spirit of this disclosure. In addition, to avoid unnecessary repetition, one or more features

shown and described in association with one embodiment may be incorporated into other embodiments unless specifically described otherwise or if the one or more features would make an embodiment non-functional.

(16) In some instances well known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

(17) This disclosure describes various instruments and portions of instruments in terms of their state in three-dimensional space. As used herein, the term “position” refers to the location of an object or a portion of an object in a three-dimensional space (e.g., three degrees of translational freedom along Cartesian x-, y-, and z-coordinates). As used herein, the term “orientation” refers to the rotational placement of an object or a portion of an object (three degrees of rotational freedom—e.g., roll, pitch, and yaw). As used herein, the term “pose” refers to the position of an object or a portion of an object in at least one degree of translational freedom and to the orientation of that object or portion of the object in at least one degree of rotational freedom (up to six total degrees of freedom). As used herein, the term “shape” refers to a set of poses, positions, or orientations measured along an object.

(18) Various embodiments of antenna systems with choke members are described herein. In some embodiments, the antenna systems are used for tissue ablation, causing an increase in a temperature of an anatomic target area by transmitting electromagnetic waves from the antenna system to the anatomic target area, or ablation site. In some embodiments, antenna systems may be flexible and suitable for use in, for example, surgical, diagnostic, therapeutic, ablative, and/or biopsy procedures. In some embodiments, antenna systems may include choke members to control current flow and localized power application. In some embodiments, the antenna systems may be used as a medical instrument in an image-guided medical procedure performed with a teleoperated medical system as described in further detail below. While some embodiments are provided herein with respect to such procedures, any reference to medical or surgical instruments and medical or surgical methods is non-limiting. In some embodiments, the antenna systems may be used for non-teleoperational procedures involving traditional manually operated medical instruments. The systems, instruments, and methods described herein may be used for animals, human cadavers, animal cadavers, portions of human or animal anatomy, non-surgical diagnosis, as well as for industrial systems and general robotic, general teleoperational, or robotic medical systems.

(19) As shown in FIGS. 1A and 1B, an antenna assembly **100** generally includes an elongate energy transmission member **102**. The elongate energy transmission member **102** includes an outer conductor **104** substantially surrounding a dielectric layer **106** and an inner conductor **108**. An insulating jacket **110** may extend over the outer conductor **104**. In this embodiment, elongate energy transmission member **102** is a coaxial cable. Other coaxial cable configurations with different configurations, shapes, etc. of inner conductor, outer conductor, and dielectric layers could also be used. In alternative embodiments, other configurations of elongate energy transmission members may be used. The antenna assembly **100** also includes a choke member **112** including a choke body **114** extending around and concentric with a portion of the elongate transmission member **102**. The choke body **114** may be formed from a tubular section of metal, a tubular mesh structure, a helical metal structure, or other types of conductive structures concentric with the energy transmission member. The choke member **112** also includes a choke connector **116** with connector components **118** electrically coupling the choke body **114** to the outer conductor **104**, through the jacket **110**. A network of passages **119** extend through spaces between the connector components **118**. In addition to providing a uniform current flow to the choke body **114**, the spaced apart connector components **118** provide flexibility to the antenna assembly **100**. The choke member **112** may be used to control current path and electromagnetic wave pattern generated by the antenna assembly **100**. More specifically, the choke member may function to cancel currents flowing on the outside of the outer conductor **104** which may lead to excess heating of the antenna assembly. As will be described in greater detail, the network of passages **119** create a fluid delivery

path, allowing a cooling fluid to pass through the choke member **112** to control the heat generated by the antenna assembly **100**.

(20) As shown in FIG. 2A, the antenna assembly **100** may be part of an antenna system **120**. In this embodiment, an antenna body **122** is coupled to the antenna assembly **100**. More specifically, the antenna body **122** may be electrically coupled (e.g., soldered) to the inner conductor **108** or the outer conductor **104**. The antenna body **122** may be used to radiate microwave energy for use in the tissue ablation process. More specifically, antenna body **122** is used to create electromagnetic radiation within a wavelength range of one meter to one millimeter, and within a frequency range of approximately 300 Megahertz (MHz) to 300 Gigahertz (GHz) (e.g., a microwave). A microwave, which is a type of radio wave, is made up of a magnetic field at a right angle to an electric field, and both the magnetic field and the electric field oscillate at a specific frequency and travel together along a direction that is perpendicular to both the magnetic field and the electric field. The microwave radiation from the combined antenna assembly **100** and antenna body **122** maybe modified and controlled by the choke member **112** to cause a desired type of ablation at an ablation target site.

(21) Antenna assembly **100** may be disposed within a flexible tubular member **124** that extends over the energy transmission member **102**, the choke member **112**, and optionally over the antenna body **122**, forming a delivery passage **128**. The tubular member **124** is open at a distal end near the antenna body **122**. Antenna body **122**, antenna assembly **100**, and tubular member **124** are disposed within a sheath **126**. In this embodiment the sheath **126** is closed, sealed, or otherwise restricts fluid from passing outside of the sheath. A return path or passage **129** extends between the tubular member **124** and the sheath **126**.

(22) The antenna system **120** also includes a fluid cooling system **130** in fluid communication with the antenna assembly **100**. Fluid cooling system **130** includes a fluid source **132** containing a cooling fluid which may be a liquid or a gaseous material. The cooling fluid may be chosen, based on its dielectric constant, to efficiently operate the antenna system. For example, water or other high dielectric constant fluids may be used to reduce a length dimension of the antenna system. Fluid cooling system **130** also includes a supply conduit **134** coupled between fluid source **132** and delivery passage **128** and a return conduit **136** coupled in fluid communication between fluid source **132** and the return passage **129**. Fluid cooling system **130** may also include pumps, valves, reservoirs or other components not shown in detail. Some or all of the components of the fluid cooling system **130** may be located outside a patient anatomy when the antenna system **120** is used for tissue ablation. The supply conduit **134** supplies a flow **138** of the fluid from fluid source **132** to the passage **128** to control the temperature along antenna assembly **100** and antenna body **122**, thereby minimizing the risk of overheating patient tissue at the target site and/or reducing distortion of a heating zone during the ablation process. For example, the fluid cooling system may be used to prevent charring of target tissue due to an over-temperature situation of the target tissue at the target site. An over-temperature situation may result from conductive and/or radiative heating of elements of antenna system **120** and/or the target tissue during operation of antenna system **120**.

(23) In this embodiment, fluid cooling system **130** is a closed loop system. Supply conduit **134** may be coupled in fluid communication with the passage **128** such that cooling fluid is supplied along the energy transmission member **102** and through the choke member **112**. The fluid is discharged within sheath **126** after exiting the tubular member **124**. The discharged fluid may then be caused to flow through the return passage **129** to return conduit **136**. The fluid flow may be generated, for example, by a pressure differential between supply conduit **134** and return conduit **136**.

(24) FIG. 2B illustrates an antenna system **140** with an open loop fluid cooling system. Some components of the system **140** are the same or substantially similar to previously described components and are identified with the same numerical identifier. In this embodiment, a fluid cooling system **142** is an open-loop fluid cooling system. Fluid cooling system **142** supplies cooling fluid through supply conduit **134** to delivery passage **128** and finally through one or more

outlet ports **144** extending through a sheath **143**. Outlet ports **144** are positioned to discharge the cooling fluid from sheath **143** into the patient anatomy or into a catheter device in which the antenna system **140** may be extended. In some embodiments, the cooling fluid discharged through outlet ports **144** may be used to flush and/or cool the ablation target site. Fluid cooling system **142** may be utilized, for example, in the open-loop configuration where bathing an ablation site in the cooling fluid is beneficial to the ablation process. In this embodiment the outlet ports are near a distal end of sheath **143**, but in alternative embodiments, outlet ports may be located at any position along the surface of the sheath, including through a distal end.

(25) FIG. 2C illustrates an antenna system **150** with a closed loop fluid cooling system. Some components of the system **150** are the same or substantially similar to previously described components and are identified with the same numerical identifiers. In this embodiment, a fluid cooling system **152** is a closed-loop fluid cooling system. Fluid cooling system **152** supplies cooling fluid through supply conduit **134** to a delivery tube **154** that extends parallel to the outer conductor **104** within the sheath **126**. In this embodiment, the delivery tube **154** terminates near the proximal end of the choke member **112**. Fluid flows through the delivery tube **154**, dispenses through the choke member **112**, and is returned to the fluid source **132** via the return passage **129**. The fluid flow may be generated, for example, by a pressure differential between supply conduit **134** and return conduit **136**.

(26) FIG. 3 illustrates a partial view of an antenna assembly **160** with a choke member **162** concentric with outer conductor **104**. The antenna assembly **160** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **162** includes a choke body **166** electrically coupled to the outer conductor **104** by a choke connector comprising a plurality of conductive beads **164**. The conductive beads may be spherical, hemispherical, or any other raised shape. In one example, the conductive beads may be solder bumps or dots. The conductive beads **164** may be arranged in rows perpendicular to a longitudinal axis A of the energy transmission member **102**, columns parallel to the longitudinal axis A, spirals around the axis A, or other patterns around the circumference of the outer conductor **104**. Spaces **168** extend between the beads **164** forming a delivery path allowing cooling fluid from a fluid cooling system to flow into direct contact with the beads and between the choke body **166** and the outer conductor **104**. The arrangement of the beads **164** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **166** generally uniform.

(27) FIG. 4 illustrates a partial view of an antenna assembly **170** with a choke member **172** concentric with outer conductor **104**. The antenna assembly **170** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **172** includes a choke body **176** electrically coupled to the outer conductor **104** by a choke connector comprising a plurality of conductive beads **174**. The conductive beads **174** may be fixed to a flexible medium **180** such as a sheet, strip, film, or other matrix that allows the conductive beads to be prearranged before flexible medium **180** is attached to the outer conductor **104**. In one example, the flexible medium may be a polymer strip that allows conductive beads **174** to be wrapped around the outer conductor **104** in a helical pattern. The conductive beads may be spherical, hemispherical, or any other raised shape. In one example, the conductive beads may be solder bumps or dots. Spaces **178** extend between the beads **174** forming a delivery path allowing cooling fluid from a fluid cooling system to flow into direct contact with the beads and between the choke body **176** and the outer conductor **104**. The arrangement of the beads **174** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **176** generally uniform.

(28) FIG. 5 illustrates a partial view of an antenna assembly **190** with a choke member **192** concentric with outer conductor **104**. The antenna assembly **190** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **192** includes a choke body **196** electrically coupled to the outer conductor **104** by a choke connector comprising a plurality of raised curved projections **194**. The curved projections **194** may be arranged in rows perpendicular

to a longitudinal axis A of the energy transmission member **102** or in other patterns around the circumference of the outer conductor **104**. The curved projections **194** may be generally arcuate in shape and curved around the axis A. In other embodiments, the curved projections may have an undulating shape with a plurality of curves around the axis A. Each of the curved projections **194** may extend less than 360 degrees around the outer conductor **104**. The ends **198** of the curved projections **194** may be parallel to the axis A or may be skewed or curved relative to the axis A. Spaces **199** extend between the curved projections **194** forming a delivery path allowing cooling fluid from a fluid cooling system to flow into direct contact with the curved projections and between the choke body **196** and the outer conductor **104**. The arrangement of the curved projections **194** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **196** generally uniform.

(29) FIG. **6** illustrates a partial view of an antenna assembly **200** with a choke member **202** concentric with outer conductor **104**. The antenna assembly **200** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **202** includes a choke body **206** electrically coupled to the outer conductor **104** by a choke connector comprising a plurality of raised curved projections **204**. The curved projections **204** may be arranged in a spiral pattern or in a multi-row spiral pattern around the longitudinal axis A of the energy transmission member **102**. Each of the curved projections **204** may extend less than 360 degrees around the outer conductor **104**. Spaces **208** extend between the curved projections **204** forming a delivery path allowing cooling fluid from a fluid cooling system to flow into direct contact with the curved projections and between the choke body **206** and the outer conductor **104**. The arrangement of the curved projections **204** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **206** generally uniform.

(30) FIG. **7** illustrates a partial view of an antenna assembly **220** with a choke member **222** concentric with outer conductor **104**. The antenna assembly **220** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **222** includes a choke body **226** electrically coupled to the outer conductor **104** by a choke connector comprising a plurality of electrically conductive tubes **224**. The tubes **224** may be arranged in a concentric pattern parallel to the axis A or, in alternative embodiments, may be wrapped around the outer conductor **104** in a spiral pattern. Cooling fluid from the fluid cooling system may flow into direct contact with the tubes **224**, either through fluid delivery paths formed by lumens **223** of the tubes **224** and/or through the spaces **199** between the tubes **224** and between the choke body **206** and the outer conductor **104**. The tubes **224** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **206** generally uniform. In various embodiments, solid rods may replace the tubes **224** with cooling fluid flowing in the spaces **199** around the rods.

(31) FIG. **8** illustrates a partial view of an antenna assembly **240** with a choke member **242** concentric with outer conductor **104**. The antenna assembly **240** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **242** includes a choke body **246** electrically coupled to the outer conductor **104**. In this embodiment, the choke member **242** may be formed from an open cell metal foam material that provides electrical conductivity from the outer conductor **104** through the sponge-like choke member **242**. The open cell choke member **242** may be generally tubular in shape and concentric with axis A. The choke member **242** may extend between the outer conductor **104** and the sheath **126**. Cooling fluid from the fluid cooling system may flow through the pores of the open cell choke member **242** which form fluid delivery paths through the choke member **242**. The open cell conductive material may provide more flexibility than a solid metal member.

(32) FIG. **9** illustrates a partial view of an antenna assembly **260** with a choke member **262** concentric with outer conductor **104**. The antenna assembly **260** may be used within a fluid-cooled assembly system (e.g., system **120**, **140**, **150**). The choke member **262** includes a choke body **266** electrically coupled to the outer conductor **104** by a choke connector comprising a conductive mesh

member **264**. The choke body **266** may be formed from a tubular section of metal, a tubular mesh structure, a helical metal structure, or other types of conductive structures concentric with the energy transmission member, as previously described. The conductive mesh member **264** may be arranged concentrically around axis A, between the choke body **266** and the outer conductor **104**. The mesh forms fluid delivery paths such that cooling fluid from the fluid cooling system may flow through the openings in the mesh member **264** between the choke body **206** and the outer conductor **104**. The mesh member **264** may have perforated openings that are shaped as squares or any other suitable shape. In some embodiments, the mesh member may be braided wire or other conductive strands. The mesh member **264** may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **266** generally uniform.

(33) FIG. **10** illustrates a method **300** for forming an antenna choke member according to some embodiments. The method **300** is illustrated as a set of operations or processes. Not all of the illustrated processes may be performed in all embodiments of method **300**. Additionally, one or more processes that are not expressly illustrated in FIG. **10** may be included before, after, in between, or as part of the processes. In some embodiments, one or more of the processes may be implemented, at least in part, in the form of executable code stored on non-transitory, tangible, machine-readable media that when run by one or more processors (e.g., the processors of a control system) may cause the one or more processors to perform one or more of the processes. In one or more embodiments, the processes may be performed by a control system (e.g., control system **412**).

(34) At a process **301** and with reference to FIG. **11A**, an outer conductor is formed by braiding a braided outer conductor layer **312a** over an outer conductor layer **312b**. The outer conductor formed of layers **312a/312b** are formed over an inner conductor **314** and a dielectric layer **316** to form an energy transmission member **310**. The braiding may be performed by a braiding machine using conductive metallic strands to form a mesh braid. The braiding machine may rotate and translate while applying wires to form the mesh. In one example, the braiding may initially progress in a direction **D1** along the longitudinal axis B to form the outer conductor layer in a single layer. In another example, the outer conductor layer can be formed of multiple layers of wire mesh by altering the direction of the braiding.

(35) At a process **302**, a spacer **318** may be inserted over the braided outer conductor layer **312a** to serve as a form for creating a choke member **320** as shown in FIG. **11B**. At a process **303**, the spacer **318** is braided over in the direction **D1** to form a choke member **320** as shown in FIG. **11C**. At a process **304**, the spacer **318** is braided over in the direction **D2**. Thus, a two layer braid formed over the spacer **318**. Alternatively, more layers may be achieved by continuing the braiding in alternating directions. At a process **305**, the spacer **318** may be removed having created a choke member **320** including a choke body **322** and a choke connector **324** coupling the choke body and the outer conductor layer **312a**. With the spacer **318** removed, the choke body **322** is spaced apart from the outer conductor **312a/312b** by a gap such that the choke body forms a cup-shaped structure around a distal portion of the outer conductor **312a/312b**. Optionally, the entire braided structure may be laminated with a lamination material such as Teflon or may be overlayed with a jacket such as a Teflon jacket.

(36) As shown in FIG. **11D**, a tubular member **326** may be formed along the energy transmission member **310** and over the choke member **320**. The tubular member **326** may be substantially similar to tubular member **124**. The tubular member **326** may extend with a sheath **328** which may be substantially similar to sheath **126**. Cooling fluid from the fluid cooling system (e.g., systems **130, 142, 152**) may flow **330** within the tubular member **326** through the braided choke connector **324** and choke body **322**. A return flow **332** of the cooling fluid is returned through the return passage between the tubular member **326** and the sheath **328**. Because the choke member **320** is created from braid, the choke member **320** allows for passage of fluid flow through the choke member **320** and in contact with the choke member **320**. The braided choke member **320** is formed from the same braid forming the outer conductor layer **312a** so provides for electrical coupling

between the outer conductor layer **312a** and the choke body **322**, while also being uniformly created during the braiding process which may minimize blockage of the flow of electrical current, keeping the current flow to the choke body **322** generally uniform.

(37) As described, the outer conductor layer **312a** and the choke member **320** may be formed of a continuous braided material. In alternative embodiments, the outer conductor may be braided separately from the mesh choke member **320** and the two may be connected (e.g., soldered) together. In alternative embodiments, the spacer may be omitted in the formation of the choke member and instead an end of the outer conductor mesh may be teased or fanned out to form the choke member. In alternative embodiments, the braided choke member may optionally be sealed at a distal end with a sealant **319**. In this sealed choke embodiment, the sheath may extend to the proximal end of the choke so that cooling fluid flows in through the tubular member **326**, dispenses into the sealed choke member and returns through the return passage between the tubular member **326** and the sheath **328**. In alternative embodiments, the first braid layer may be formed separately (i.e. not continuously) from the second braid layer and the two layers may be welded together.

(38) FIG. **12** illustrates an alternative choke member **340** used with energy transmission member **310**. In this embodiment, after the braided outer conductor is formed (process **301**) and before the spacer is inserted over the outer conductor (process **302**), a set of pins **342** are wrapped around and connected to the outer conductor layer **312a**. The pins **342** will pierce any jacket or protective coating extending over the outer conductor layer **312a**. In some embodiments the pins **342** may be initially coupled to a flexible strip that is wrapped around the outer conductor. After the pins are connected to the outer conductor layer **312a**, the flexible strip may be removed from the set of pins **342** allowing the pins to remain coupled to the outer conductor. After the pins **342** are in place, the method **300** may continue by inserting the spacer adjacent to the pins, and the spacer and pins may be braided over in multiple directions. The pins **342** form part of the choke connector such that cooling fluid from the fluid cooling system may flow between the pins and through the braided mesh choke connector **324** between the choke body **322** and the outer conductor layer **312a**.

(39) FIG. **13** illustrates an alternative choke member **350** used with energy transmission member **310**. In this embodiment, after the braided outer conductor layer **312a** is formed (process **301**) a band is etched into the outer conductor and a conductive ring member **352** is placed in the etched band, thus electrically coupling the ring member **352** to the outer conductor layer **312a** as a choke connector. A portion of a hypotube **354** may form a choke body and may be placed around the energy transmission member **310**, adjacent to the ring member **352**. The hypotube **354** and the ring member **352** may be braided over to form the choke member **350**.

(40) FIG. **14** illustrates a partial view of an antenna assembly **360** with a choke member **362** concentric with an outer conductor **364**. The outer conductor **364** and the choke member **362** may be formed of a braided mesh material. In this embodiment, a sealant layer **366** may be applied to the braided mesh material to minimize the ingress of fluid that may degrade antenna performance. The sealant layer **366** may be painted, sputtered, evaporated, deposited, or otherwise applied to the braided mesh.

(41) FIG. **15** illustrates a partial view of an antenna assembly **370** with a choke member **372** concentric with an outer conductor **374**. The outer conductor **374** and the choke member **372** may be formed of a braided mesh material. In this embodiment, a sealed cap **376** is clamped around the outer conductor **374** to limit water ingress that may degrade antenna performance.

(42) FIG. **16** illustrates a method **380** for transferring energy to an ablation site according to some embodiments. The method **380** is illustrated as a set of operations or processes. Not all of the illustrated processes may be performed in all embodiments of method **380**. Additionally, one or more processes that are not expressly illustrated in FIG. **16** may be included before, after, in between, or as part of the processes. In some embodiments, one or more of the processes may be implemented, at least in part, in the form of executable code stored on non-transitory, tangible, machine-readable media that when run by one or more processors (e.g., the processors of a control

system) may cause the one or more processors to perform one or more of the processes. In one or more embodiments, the processes may be performed by a control system (e.g., control system **412**). (43) At a process **382**, energy is conducted through an energy transmission member (e.g. energy transmission member **102**). At a process **384**, energy is conducted through a choke member (e.g., choke member **112**) coupled to the energy transmission member. As described in the preceding embodiments, the choke member may include a choke body and a choke connector electrically coupling the choke body to the energy transmission member. At a process **386**, cooling fluid is provided from a cooling system through the choke member. The fluid may directly contact the choke connector. At an optional process **388**, the cooling fluid is returned to the cooling system through a sheath extending over the energy transmission member. Alternatively, at an optional process **390**, the cooling fluid may be discharged through a port in the sheath extending over the energy transmission member.

(44) As shown in FIG. **17**, medical system **400** generally includes a manipulator assembly **402** for operating a medical instrument **404** in performing various procedures on a patient P positioned on a table T. In some embodiments, the medical instrument **404** may include any of the antenna instruments described herein, where the antenna instruments are integrated into a distal end portion of the medical instrument **404**. In alternative embodiments, the medical instrument **404** may deliver any of the antenna instruments described herein. The manipulator assembly **402** may be teleoperated, non-teleoperated, or a hybrid teleoperated and non-teleoperated assembly with select degrees of freedom of motion that may be motorized and/or teleoperated and select degrees of freedom of motion that may be non-motorized and/or non-teleoperated. Master assembly **406** generally includes one or more control devices for controlling manipulator assembly **402**. Manipulator assembly **402** supports medical instrument **404** and may optionally include a plurality of actuators or motors that drive inputs on medical instrument **404** in response to commands from a control system **412**. The actuators may optionally include drive systems that when coupled to medical instrument **404** may advance medical instrument **404** into a naturally or surgically created anatomic orifice. Other drive systems may move the distal end of medical instrument **04** in multiple degrees of freedom, which may include three degrees of linear motion (e.g., linear motion along the X, Y, Z Cartesian axes) and in three degrees of rotational motion (e.g., rotation about the X, Y, Z Cartesian axes). Additionally, the actuators can be used to actuate an articulable end effector of medical instrument **404** for grasping tissue in the jaws of a biopsy device and/or the like. Actuator position sensors such as resolvers, encoders, potentiometers, and other mechanisms may provide sensor data to medical system **400** describing the rotation and orientation of the motor shafts. This position sensor data may be used to determine motion of the objects manipulated by the actuators.

(45) Teleoperated medical system **400** also includes a display system **410** for displaying an image or representation of the surgical site and medical instrument **404** generated by sub-systems of sensor system **408** and/or any auxiliary information related to a procedure including information related to ablation (e.g., temperature, impedance, energy delivery power levels, frequency, current, energy delivery duration, indicators of tissue ablation, etc.). Display system **410** and master assembly **406** may be oriented so operator O can control medical instrument **404** and master assembly **406** with the perception of telepresence.

(46) In some embodiments, medical instrument **404** may include components of an imaging system, which may include an imaging scope assembly or imaging instrument that records a concurrent or real-time image of a surgical site and provides the image to the operator or operator O through one or more displays of medical system **400**, such as one or more displays of display system **410**. The concurrent image may be, for example, a two or three-dimensional image captured by an imaging instrument positioned within the surgical site. In some embodiments, the imaging system includes endoscopic imaging instrument components that may be integrally or removably coupled to medical instrument **404**. However, in some embodiments, a separate endoscope,

attached to a separate manipulator assembly may be used with medical instrument **404** to image the surgical site. In some embodiments, the imaging system includes a channel (not shown) that may provide for a delivery of instruments, devices, catheters, and/or antenna instruments described herein. The imaging system may be implemented as hardware, firmware, software or a combination thereof which interact with or are otherwise executed by one or more computer processors, which may include the processors of the control system **412**.

(47) Teleoperated medical system **400** may also include control system **412**. Control system **412** includes at least one memory and at least one computer processor (not shown) for effecting control between medical instrument **404**, master assembly **406**, sensor system **408**, and display system **410**. Control system **412** also includes programmed instructions (e.g., a non-transitory machine-readable medium storing the instructions) to implement some or all of the methods described in accordance with aspects disclosed herein, including instructions for providing information to display system **410**.

(48) Control system **412** may optionally further include a virtual visualization system to provide navigation assistance to operator **O** when controlling medical instrument **404** during an image-guided surgical procedure. Virtual navigation using the virtual visualization system may be based upon reference to an acquired preoperative or intraoperative dataset of anatomic passageways. The virtual visualization system processes images of the surgical site imaged using imaging technology such as computerized tomography (CT), magnetic resonance imaging (Mt), fluoroscopy-, thermography, ultrasound, optical coherence tomography (OCT), thermal imaging, impedance imaging, laser imaging, nanotube X-ray imaging, and/or the like.

(49) FIG. **18A** is a simplified diagram of a medical instrument system **500** according to some embodiments. Medical instrument system **500** includes elongate device **502**, such as a flexible catheter, coupled to a drive unit **504**. Elongate device **502** includes a flexible body **516** having proximal end **517** and distal end or tip portion **518**. The distal end **518** may include any of the antenna instruments described herein. Medical instrument system **500** further includes a tracking system **530** for determining the position, orientation, speed, velocity, pose, and/or shape of distal end **518** and/or of one or more segments **524** along flexible body **516** using one or more sensors and/or imaging devices as described in further detail below.

(50) Tracking system **530** may optionally track distal end **518** and/or one or more of the segments **524** using a shape sensor **522**. Shape sensor **522** may optionally include an optical fiber aligned with flexible body **516** (e.g., provided within an interior channel (not shown) or mounted externally). The optical fiber of shape sensor **522** forms a fiber optic bend sensor for determining the shape of flexible body **516**. In one alternative, optical fibers including Fiber Bragg Gratings (FBGs) are used to provide strain measurements in structures in one or more dimensions. Various systems and methods for monitoring the shape and relative position of an optical fiber in three dimensions are described in U.S. patent application Ser. No. 11/180,389 (filed Jul. 13, 2005) (disclosing “Fiber optic position and shape sensing device and method relating thereto”); U.S. patent application Ser. No. 12/047,056 (filed on Jul. 16, 2004) (disclosing “Fiber-optic shape and relative position sensing”); and U.S. Pat. No. 6,389,187 (filed on Jun. 17, 1998) (disclosing “Optical Fibre Bend Sensor”), which are all incorporated by reference herein in their entireties. In some embodiments, tracking system **530** may optionally and/or additionally track distal end **518** using a position sensor system **520**. Position sensor system **520** may be a component of an EM sensor system with position sensor system **520** including one or more conductive coils that may be subjected to an externally generated electromagnetic field. In some embodiments, position sensor system **520** may be configured and positioned to measure six degrees of freedom, e.g., three position coordinates X, Y, Z and three orientation angles indicating pitch, yaw, and roll of a base point or five degrees of freedom, e.g., three position coordinates X, Y, Z and two orientation angles indicating pitch and yaw of a base point. Further description of a position sensor system is provided in U.S. Pat. No. 6,380,732 (filed Aug. 11, 1999) (disclosing “Six-Degree of Freedom Tracking

System Having a Passive Transponder on the Object Being Tracked”), which is incorporated by reference herein in its entirety. In some embodiments, an optical fiber sensor may be used to measure temperature or force. In some embodiments, a temperature sensor, a force sensor, an impedance sensor, or other types of sensors may be included within the flexible body.

(51) Flexible body **516** includes a channel **521** sized and shaped to receive a medical instrument **526**. In various embodiments, any of the antenna instruments and sheaths described above may be inserted through the channel **521** of the flexible body **516**. FIG. **18B** is a simplified diagram of flexible body **516** with medical instrument **526** extended according to some embodiments. In some embodiments, medical instrument **526** may be used for procedures such as imaging, visualization, surgery, biopsy, ablation, illumination, irrigation, or suction. Medical instrument **526** can be deployed through channel **521** of flexible body **516** and used at a target location within the anatomy. Medical instrument **526** may include, for example, image capture probes, biopsy instruments, laser ablation fibers, and/or other surgical, diagnostic, or therapeutic tools. Medical instrument **526** may be used with an imaging instrument (e.g., an image capture probe) also within flexible body **516**. The imaging instrument may include a cable coupled to the camera for transmitting the captured image data. In some examples, the imaging instrument may be a fiber-optic bundle, such as a fiberscope, that couples to image processing system **531**. The imaging instrument may be single or multi-spectral, for example capturing image data in one or more of the visible, infrared, and/or ultraviolet spectrums. Medical instrument **526** may be advanced from the opening of channel **521** to perform the procedure and then retracted back into the channel when the procedure is complete. Medical instrument **526** may be removed from proximal end **517** of flexible body **516** or from another optional instrument port (not shown) along flexible body **516**.

(52) Flexible body **516** may also house cables, linkages, or other steering controls (not shown) that extend between drive unit **504** and distal end **518** to controllably bend distal end **518** as shown, for example, by broken dashed line depictions **519** of distal end **518**. In some examples, at least four cables are used to provide independent “up-down” steering to control a pitch of distal end **518** and “left-right” steering to control a yaw of distal end **518**. Steerable elongate devices are described in detail in U.S. patent application Ser. No. 13/274,208 (filed Oct. 14, 2011) (disclosing “Catheter with Removable Vision Probe”), which is incorporated by reference herein in its entirety.

(53) The information from tracking system **530** may be sent to a navigation system **532** where it is combined with information from image processing system **531** and/or the preoperatively obtained models to provide the operator with real-time position information. In some examples, the real-time position information may be displayed on display system **410** of FIG. **17** for use in the control of medical instrument system **500**. In some examples, control system **412** of FIG. **17** may utilize the position information as feedback for positioning medical instrument system **500**. Various systems for using fiber optic sensors to register and display a surgical instrument with surgical images are provided in U.S. patent application Ser. No. 13/107,562, filed May 13, 2011, disclosing, “Medical System Providing Dynamic Registration of a Model of an Anatomic Structure for Image-Guided Surgery,” which is incorporated by reference herein in its entirety.

(54) In some examples, medical instrument system **500** may be teleoperated within medical system **400** of FIG. **17**. In some embodiments, manipulator assembly **402** of FIG. **17** may be replaced by direct operator control. In some examples, the direct operator control may include various handles and operator interfaces for hand-held operation of the instrument.

(55) One or more elements in embodiments of this disclosure may be implemented in software to execute on a processor of a computer system such as control processing system. When implemented in software, the elements of the embodiments of the invention are essentially the code segments to perform the necessary tasks. The program or code segments can be stored in a processor readable storage medium or device that may have been downloaded by way of a computer data signal embodied in a carrier wave over a transmission medium or a communication link. The processor readable storage device may include any medium that can store information

including an optical medium, semiconductor medium, and magnetic medium. Processor readable storage device examples include an electronic circuit; a semiconductor device, a semiconductor memory device, a read only memory (ROM), a flash memory, an erasable programmable read only memory (EPROM); a floppy diskette, a CD-ROM, an optical disk, a hard disk, or other storage device. The code segments may be downloaded via computer networks such as the Internet, Intranet, etc. Any of a wide variety of centralized or distributed data processing architectures may be employed. Programmed instructions may be implemented as a number of separate programs or subroutines, or they may be integrated into a number of other aspects of the systems described herein. In one embodiment, the control system supports wireless communication protocols such as Bluetooth, IrDA, HomeRF, IEEE 802.11, DECT, and Wireless Telemetry.

(56) Medical tools that may be delivered through the flexible elongate devices or catheters disclosed herein may include, for example, image capture probes, biopsy instruments, laser ablation fibers, and/or other surgical, diagnostic, or therapeutic tools. Medical tools may include end effectors having a single working member such as a scalpel, a blunt blade, an optical fiber, an electrode, and/or the like. Other end effectors may include, for example, forceps, graspers, scissors, clip applicators, and/or the like. Other end effectors may further include electrically activated end effectors such as electrosurgical electrodes, transducers, sensors, and/or the like. Medical tools may include image capture probes that include a stereoscopic or monoscopic camera for capturing images (including video images). Medical tools may additionally house cables, linkages, or other actuation controls (not shown) that extend between its proximal and distal ends to controllably bend the distal end of an antenna instrument such as elongate energy transmission member **102**. Steerable instruments are described in detail in U.S. Pat. No. 7,416,681 (filed on Oct. 4, 2005) (disclosing “Articulated Surgical Instrument for Performing Minimally Invasive Surgery with Enhanced Dexterity and Sensitivity”) and U.S. patent application Ser. No. 12/286,644 (filed Sep. 30, 2008) (disclosing “Passive Preload and Capstan Drive for Surgical Instruments”), which are incorporated by reference herein in their entireties.

(57) The systems described herein may be suited for navigation and treatment of anatomic tissues, via natural or surgically created connected passageway's, in any of a variety of anatomic systems, including the lung, colon, stomach, the intestines, the kidneys and kidney calices, bladder, liver, gall bladder, pancreas, spleen, the ureter, ovaries, uterus, the brain, the heart, the circulatory system including vasculature, and/or the like.

(58) Note that the processes and displays presented may not inherently be related to any particular computer or other apparatus. Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the operations described. The required structure for a variety of these systems will appear as elements in the claims. In addition, the embodiments of the invention are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the invention as described herein.

(59) While certain exemplary embodiments of the invention have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of and not restrictive on the broad invention, and that the embodiments of the invention not be limited to the specific constructions and arrangements shown and described, since various other modifications may occur to those ordinarily skilled in the art.

Claims

1. An antenna system for tissue ablation, the antenna system comprising: an energy transmission member; an antenna body coupled to the energy transmission member; a fluid source; and a choke member including a choke body and a choke connector electrically coupling the choke body to the

energy transmission member, the choke connector in direct contact with fluid from the fluid source and forming a delivery path for the fluid between the choke body and the energy transmission member, wherein the choke connector comprises a plurality of conductive beads extending between the energy transmission member and the choke body, wherein each conductive bead of the plurality of conductive beads is coupled to the choke body independent of each other conductive bead of the plurality of conductive beads.

2. The antenna system of claim 1 wherein the choke body is concentric with the energy transmission member.

3. The antenna system of claim 1 wherein the choke body is a tubular member.

4. The antenna system of claim 1 wherein the choke connector is concentric with the energy transmission member.

5. The antenna system of claim 1 wherein the plurality of conductive beads are arranged in a helical pattern.

6. The antenna system of claim 1 wherein the plurality of conductive beads are coupled to a flexible sheet that is coupled to the energy transmission member.

7. The antenna system of claim 1 further comprising: a flexible tube extending over the energy transmission member and the choke member, wherein the delivery path for the fluid is between the flexible tube and the energy transmission member; and a sheath extending over the flexible tube, the sheath forming a return path for the fluid dispensed from a distal end of the flexible tube.

8. An antenna system for tissue ablation, the antenna system comprising: an energy transmission member; an antenna body coupled to the energy transmission member; a fluid source; and a choke member including a choke body and a choke connector electrically coupling the choke body to the energy transmission member, the choke connector in direct contact with fluid from the fluid source and forming a delivery path for the fluid between the choke body and the energy transmission member, wherein the choke connector includes a plurality of curved projections arranged around the energy transmission member, wherein each curved projection of the plurality of curved projections is coupled to the choke body independent of each other curved projection of the plurality of curved projections.

9. The antenna system of claim 8 wherein at least one of the plurality of curved projections has a first dimension along a longitudinal axis of the energy transmission member and a second dimension perpendicular to the longitudinal axis of the energy transmission member and wherein the second dimension is greater than the first dimension.

10. The antenna system of claim 8 wherein at least one of the plurality of the curved projections extends less than 360 degrees around the energy transmission member.

11. The antenna system of claim 8 wherein the choke body is concentric with the energy transmission member.

12. The antenna system of claim 11 wherein the choke body is a tubular member.

13. The antenna system of claim 8 further comprising: a flexible tube extending over the energy transmission member and the choke member, wherein the delivery path for the fluid is between the flexible tube and the energy transmission member; and a sheath extending over the flexible tube, the sheath forming a return path for the fluid dispensed from a distal end of the flexible tube.
